How-To Library

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Enable Proxy User Support for Oracle Databases in PowerCenter (10.4.0)

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Abstract

You can temporarily sign in to your network as a different user through user impersonation. This article describes how you can specify an impersonate user and connect to Oracle databases on behalf of that user and perform read or write operations.

Supported Versions

- PowerCenter 10.4.0

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Overview

You can temporarily sign in to your network as a different user through user impersonation. With full impersonation permissions you can impersonate all other users in their network and take any action, regardless of the impersonating user's permission level.

With a proxy user account, you can connect as another user using your own user name and password. You can act as an impersonate user with the password of the proxy user.

Use Oracle proxy user account to impersonate as a different user to access tables that has privileges granted to impersonate users. Use the `GRANT CONNECT THROUGH SQL` command on an Oracle database to proxy an impersonate user. For example, you have a main database user, `MAIN_USER` and a proxy user, `PROXY_USER`. You can grant access to `PROXY_USER` to access tables that the `MAIN_USER` owns without revealing the password of `MAIN_USER` to the `PROXY_USER`.

Perform the following steps to grant proxy privileges to the `PROXY_USER`:

1. Create a proxy user.

```
CONNECT / AS SYSDBA
CREATE USER PROXY_USER IDENTIFIED BY PROXY_PASSWD;
```

2. Grant proxy privileges to the proxy user.

```
> ALTER USER MAIN_USER GRANT CONNECT THROUGH PROXY_USER;
> QUIT;
```

3. Connect to the main user using the proxy user credentials.

```
> CONNECT PROXY_USER[MAIN_USER]/PROXY_PASSWD;
> SHOW USER
USER is "MAIN_USER"
```

Effective in version 10.4.0, you can specify the impersonate user name in the Oracle connection in PowerCenter to connect to Oracle sources and targets and transformations such as, Lookup, SQL, and Stored Procedure transformations. When you specify an impersonate user, you can connect to Oracle on behalf of that user and perform read or write operations. You can use the impersonate user at both design time and run time.

You can enable user impersonation for mappings with pushdown optimization. You can also enable bulk mode on target Oracle object when using impersonation.

Configure User Impersonation During Metadata Import

You can import table definitions from Oracle to create Oracle source or target definitions.

Create the data source on the machine where you installed the PowerCenter Client. Configure the data source in the ODBC Data Source Administrator. You must create the 32-bit data source for the PowerCenter Client. You can use the 32-bit DataDirect ODBC Oracle Wire Protocol driver to create the data source. This ODBC driver is available with the PowerCenter installation and has the capability to allow users to specify the impersonate user.

1. To create a 32-bit data source for the PowerCenter Client, open Control Panel and click **Administrative Tools**. Then, double-click **ODBC Data Sources (32-bit)**.

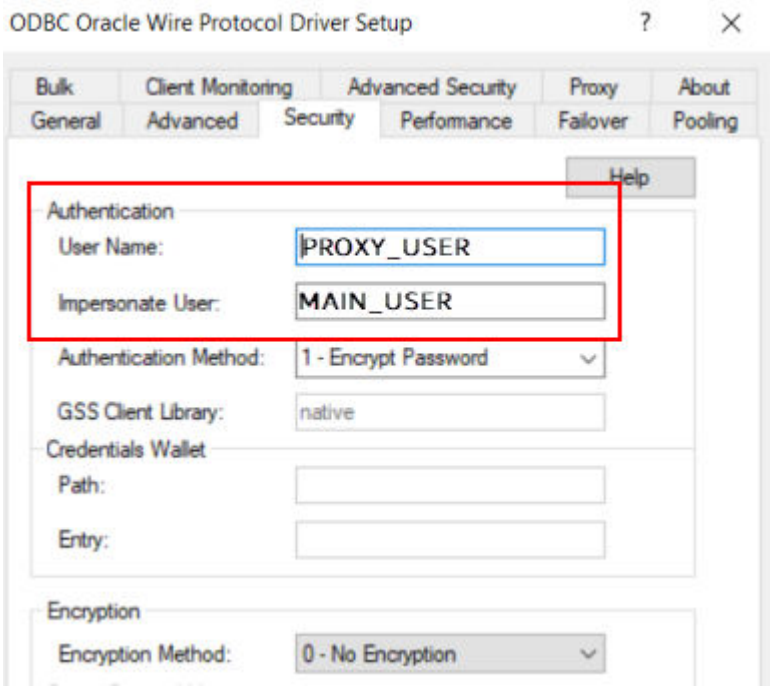
   Alternatively, enter the following command at the command prompt:

   ```
   %windir%\SysWOW64\odbcad32.exe
   ```

   The **ODBC Data Source Administrator** wizard appears.
2. On the **System DSN** tab of the ODBC Data Source Administrator wizard, click **Add**
3. In the **Create a New Data Source** dialog box, select **DataDirect 8.0 Oracle Wire Protocol** from the list of available drivers, and click **Finish**
4. On the **General** tab of the **ODBC Oracle Wire Protocol Driver Setup** dialog box, enter the required details.

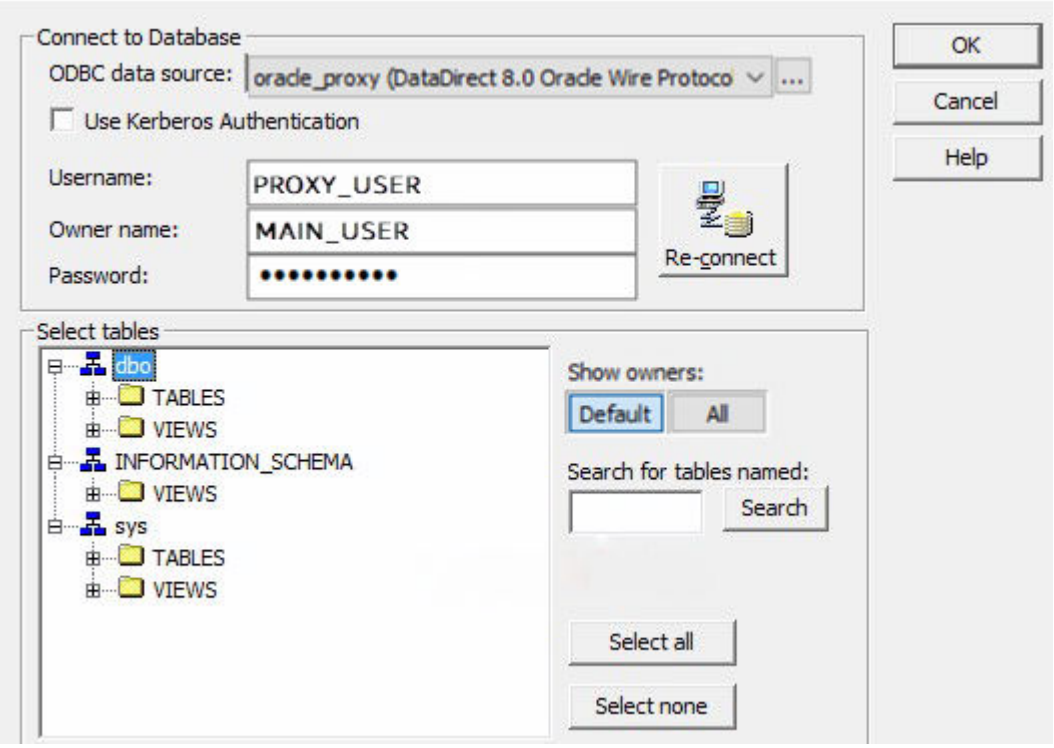
5. On the **Security** tab, configure the required encryption properties.
   Enter the user name as `PROXY_USER` and impersonate user as `MAIN_USER`.

   The following image shows the DataDirect 8.0 Oracle Wire protocol setup where you configure the security settings:



   When you configure the DSN to enable proxy user to impersonate as another user, the list of source, target, and stored procedure tables of the impersonate user appears in the PowerCenter Designer as shown in the

following image:



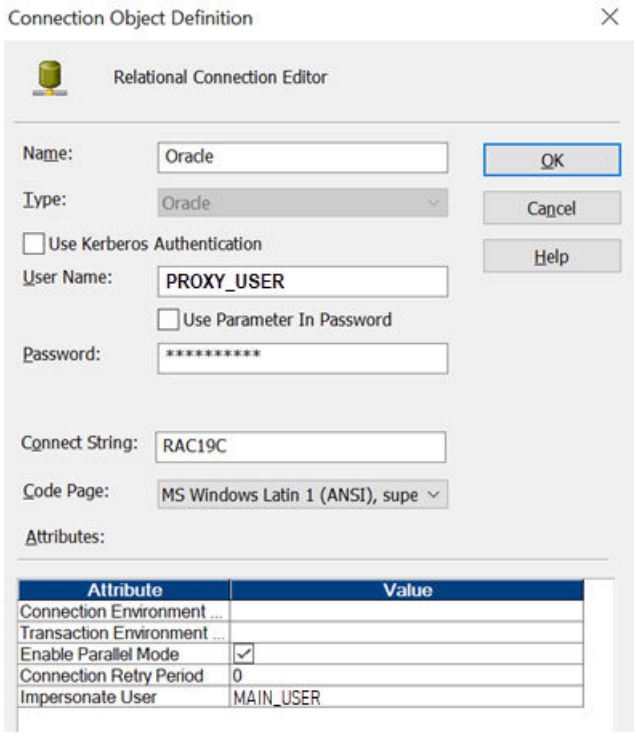
Change the **Owner name** value to the impersonate user name when **Show owners** is set to **Default**.

Setting Impersonate User At Run Time

Before you create and run sessions, you must configure connections in the Workflow Manager. A connection object is a global object that defines a connection in the repository. You create and change connection objects and assign permissions to connection objects in the Workflow Manager.

In the Workflow Manager, you can create or edit Oracle connections to enable user impersonation. Use a relational connection object for each source, target, lookup, and stored procedure database that you want to access. Enter the impersonate user name, `MAIN_USER` for the **Impersonate User** attribute in the **Relational Connection Editor** dialog box.

The following image shows the **Relational Connection Editor** dialog box with user impersonation enabled:



As in Designer, you can assign relational connection object to source, target, Lookup, Stored Procedures, or SQL transformation objects for all Oracle connections. Any unqualified table object names assume **Impersonate user** value as its schema even though the logged in user is a proxy user. In such case, if the object does not exist in the impersonated schema, the mapping fails at run time with `<Table/View/Stored Procedure> doesn't exist` error. If the proxy user tries to impersonate as a different user without the required privileges, the mapping fails at run time with appropriate error message from the Oracle database.

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